PARKISCAN NEUROVERSE

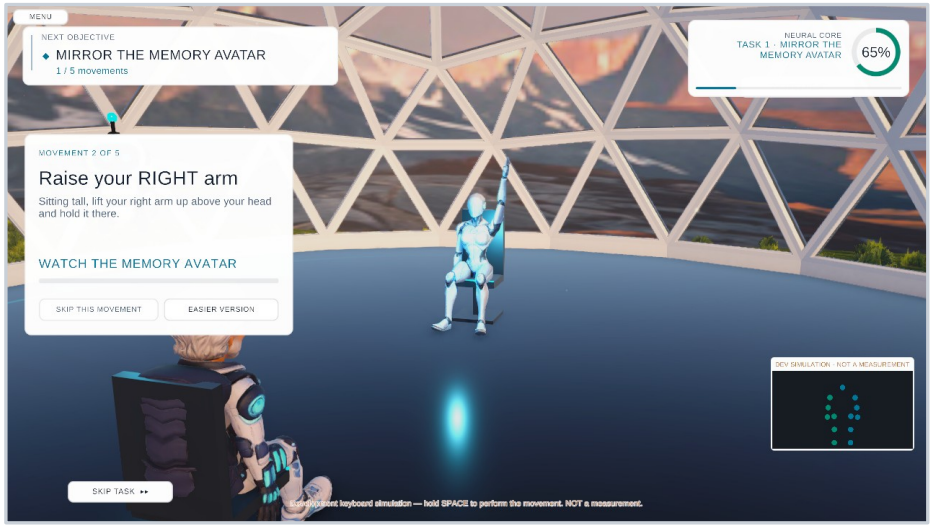
A game-based home assessment and digital twin for Parkinson’s disease

GAME · VOICE · EYES · MOTION · DIGITAL TWIN · CLINICIAN DASHBOARD

ParkiScan Neuroverse · Game + Digital Twin

Happiest Health · Reimagining and Reforming Healthcare in India Summit 2026 · Digital Twin Challenge, Phase 1

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Integrated M.Tech CSE with Business Analytics · VIT Chennai



Level 3 · Mirror the Memory Avatar (development simulation, earlier UI theme)

Three levels. One story. Eye tracking, voice, drawing, memory, reaction time, movement and gait are collected as game objectives, then land on one patient record and a clinician dashboard.

The problem, and why a game

THE PROBLEM

1. About 5.8 lakh people in India live with Parkinson's, roughly 10% of the global burden [1].

2. Early signs can be subtle. A prodromal phase with small, non-obvious changes can precede diagnosis [2], and mild changes are easy to put down to ageing.

3. Care relies on occasional clinic visits and questionnaires: a snapshot every few months, based on recall.

4. Symptoms fluctuate with medication state and happen at home, between visits. Wearable and home technology can supply objective data between visits [8].

WHY AN INTERACTIVE GAME

ESTABLISHED

Gamification is used to raise engagement in health interventions; the review finds the evidence promising but limited [3].

ESTABLISHED

Smartphone tasks (voice, tapping, gait, balance, memory) can quantify Parkinson's severity at scale [6]; eye-movement changes are documented in PD [7].

ESTABLISHED

VR / exergames are feasible and safe in PD; versus conventional exercise results were similar, with low-certainty evidence (8 trials, 263 people) [4,5].

NEUROVERSE · PROPOSED

Neuroverse turns these tasks into one 3-level story with avatars, feedback and a skip on every task, so repeat sessions at home stay tolerable. Engagement and diagnostic value are not yet tested.

Not a standalone diagnostic tool. Neuroverse records objective measures with quality flags so a clinician can compare sessions. The overall Parkinson's classification is withheld until it is validated on clinician-labelled sessions.

[1] Khurana & Gourie-Devi 2025 · [2] Postuma 2015 · [3] Johnson 2016 · [4] Dockx 2016 (Cochrane) · [5] Garcia-Agundez 2019 · [6] Zhan 2018 · [7] Anderson & MacAskill 2013 · [8] Espay 2016

Complete game flow: assessments as objectives

Each assessment is a game objective in one continuous story, never a separate test. Every task can be skipped; a skip is recorded as not collected.



LEVEL 1 · VISUAL · VOICE · MEMORY

Follow the Memory Lantern

Webcam iris gaze after a 5-point check. Pointer error, % on target, re-acquisition latency, flash reaction time.

Memory Core (microphone)

Sustained “Ahh”, pa-ta-ka, clap to 90 BPM, repeat a sentence. F0, jitter and shimmer proxies, HNR, pauses, rhythm CV, speech rate.

Recover the lost memory

Hits, omissions, false recognitions, order match, recall time.

LEVEL 2 · FINE MOTOR · COGNITION · REACTION

Trace the neural circuit (spiral)

Radial deviation, 4–7 Hz tremor-band proxy, LDLJ smoothness, pen lifts.

Connect nodes · Repair component

Sequencing accuracy, wrong links, inter-selection time, placement and rotation error.

Reactor signal

Reaction time mean and SD, false and missed responses.

LEVEL 3 · WEBCAM POSE · GAIT · SELF-REPORT

Mirror the Memory Avatar

Five arm movements: peak elevation, pose similarity, tracking confidence.

Standing path or seated drive

March on the spot: steps, cadence, step-interval SD, stop/go latency. Seated: steering wheel, lane offset, brake and go latency.

Six check-ins (questionnaire)

Self-report per domain, shown as given, never scored.

Telemetry on every task: completion time, accuracy, errors, response pattern, input source and a data-quality state (valid, caveats, insufficient, invalid, not collected).

Real-time sensors: what comes from where

Three separate sources, never mixed up: webcam and microphone, wearable IMUs, and the game itself.

WEBCAM + MICROPHONE

IMPLEMENTED

on-device, no upload

- Face and iris mesh (478 points): gaze for the lantern
- BlazePose (33 points): arm elevation, steps, wheel steering
- Microphone: voice features and clap timing
- 2-D, single camera: an estimate, not a clinical-grade measure

WRIST + ANKLE IMU

BUILT · LIMITED

ESP32 + MPU-6050 · 100 Hz · ±4 g · ±500 °/s

- Wrist: acceleration and angular velocity, 4–7 Hz tremor-band ratio, peak frequency, gyro RMS
- Ankle: cadence, step regularity, freeze index and band power (as in Model A)
- Complements the camera in spiral, gait and seated tasks

GAME TELEMETRY

IMPLEMENTED

every task, every session

- Task completion time, accuracy, errors, response latency
- Pointer and trace paths, input source, skip state
- Per-domain quality flags; QA or simulated runs are labelled

IMU DATA PATH: TIMESTAMP → TRANSMIT → VALIDATE → SYNCHRONISE → FEATURES

ESP32 + MPU-6050
seq, t_ms, t_us per sample · 921600 baud JSON

→

Python receiver
schema and range checks; duplicates, out-of-order and stale data rejected

→

Local UDP 127.0.0.1
validated samples only; simulated packets flagged

→

Unity bridge
placement-tagged; aligned to the game clock

→

Per-task window
features stored with the task record; raw readings kept as CSV

→

Session record
stored with the session; the twin's wearable grid is simulated or Daphnet today

Status. Firmware, receiver, Unity bridge, simulator and tests are built. The ankle model is trained on real Daphnet accelerometer data; the wrist and ankle boards have not yet been run through a full play session on live hardware. The wrist tremor features exist, but no wrist model is trained.

Moore-type freeze index and Daphnet data: Bächlin et al. 2010 [9] · Wearable technology in PD: Espay et al. 2016 [8]

AI, datasets and methodology

Ten datasets reviewed. Seven were used for training or evaluation, all with participant-grouped validation; three are proposed.

Dataset	Modality	Role in Neuroverse	Status
Daphnet Freezing of Gait	3 accelerometers, 10 people, 64 Hz	Model A (freeze in 3 s); 1-D CNN; twin clinical header	Trained · LOSO
Gait in Neurodegenerative Disease	Insole stride intervals, 31 people	Gait PD vs control; severity (research)	Trained · tiny set
Spiral drawings (digitised tablet)	Pen trajectories, 40 people	Spiral model, same extractor as the game	Trained · experimental
Parkinsons (Oxford voice)	Sustained-vowel features, 32 people	Voice research model, not applied to game audio	Trained · research only
Parkinson's speech recordings	26 recordings × 40 people	Speech research model	Flagged · failed held-out test
Parkinson's Telemonitoring	Home voice, UPDRS, 42 people	UPDRS regression	Flagged · below mean baseline
ET_PD_VIS	Gaze heat-map and scan-path images	Image-feature model on TMT A/B/C	Flagged · AUC 1.00, likely confound
NDDCNN	Not verified by us	Not in the repository	Proposed
Parkinson's Disease Classification	Vowel /a/ ×3, 252 people, 754 features	Candidate for voice-model evaluation	Proposed
TMT-PD	Eye-tracker events, 28 PD + 12 controls	Candidate for gaze-task validation	Proposed

Multimodal

Per-domain outputs sit side by side with quality flags. A fused classifier is proposed once clinician-labelled sessions exist.

Proposed only

Vision Transformers, LSTMs and Transformer speech encoders for diagnosis are not implemented.

No diagnostic accuracy is claimed.

MODEL FAMILIES AND THEIR ACTUAL ROLES

Gradient boosting · random forest · MLP
tabular features: Models A and B, spiral, voice, gait

1-D CNN (raw 3×256 windows)
Daphnet detection, AUC 0.85 vs 0.88 boosting

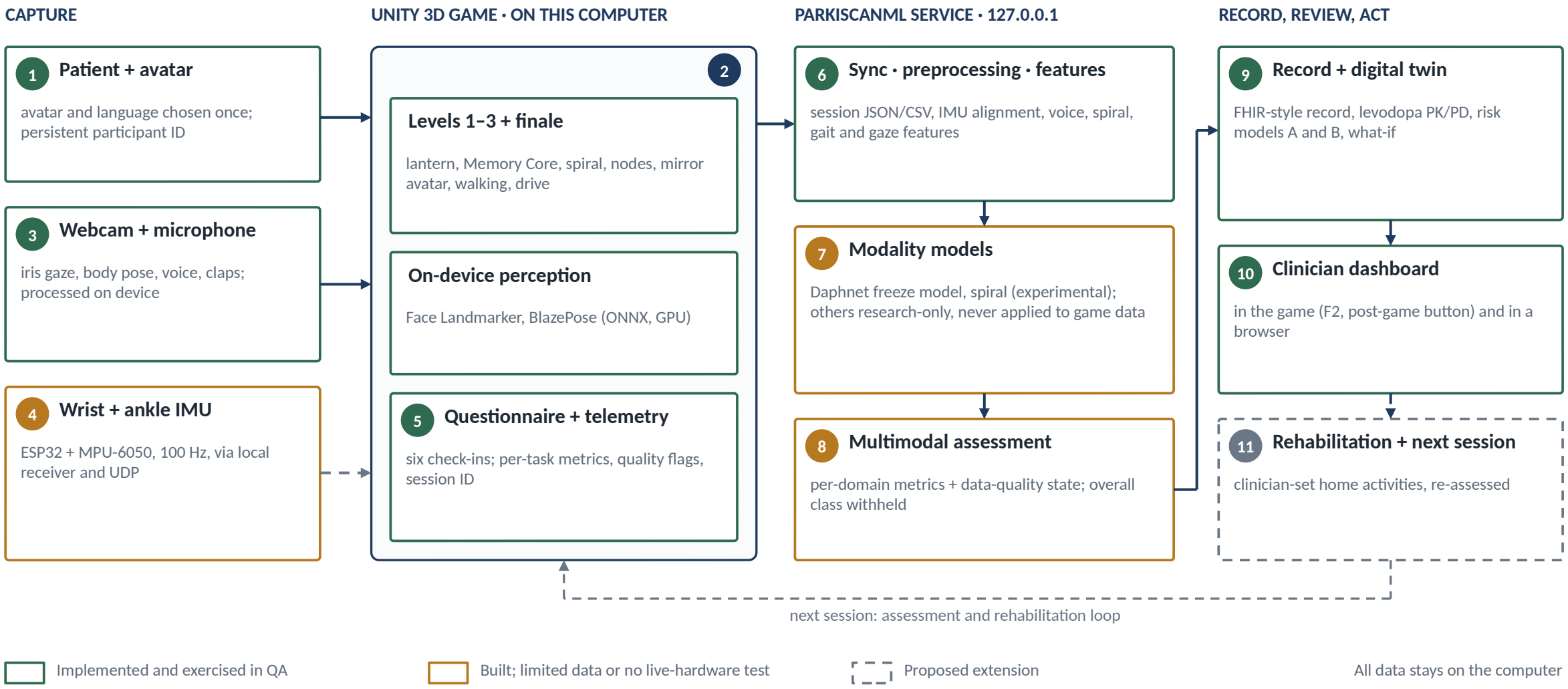
Pretrained CNN perception
BlazeFace, Face Landmarker, BlazePose: gaze and pose, not trained by us

Transformer ASR (faster-whisper, local)
follows the sentence task; not a diagnostic model

Voice biomarkers for PD telemonitoring [10] · Daphnet [9] · Datasets: Hausdorff 2000, Isenkul 2014, Little 2007, Sakar 2013/2018, Tsanas 2010, Mendeley Data ET_PD_VIS and TMT-PD (2026)

System architecture

Capture on the device, process locally, build one patient record, review with a clinician, feed the next session.



Avatar, patient record and the digital twin

AVATAR AND PATIENT IDENTITY

- The player picks an avatar and outfit at the start.
- A persistent participant ID is created on this device and stamped on every task, session and telemetry event.
- Sessions accumulate under that ID: one longitudinal record.
- The avatar is the player's visual self in the game. The twin is the data: it is keyed by the participant ID, not by the avatar.

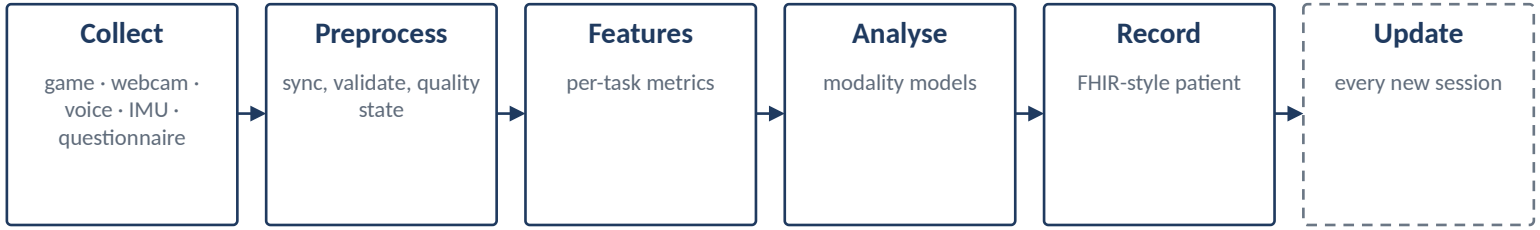
TWIN VERSUS PROFILE VERSUS DASHBOARD

Static profile a snapshot of facts

Dashboard displays data, does not model it

Digital twin one patient's record plus models, updated by each session; simulates and forecasts

TWIN CREATION PIPELINE



IMPLEMENTED NOW

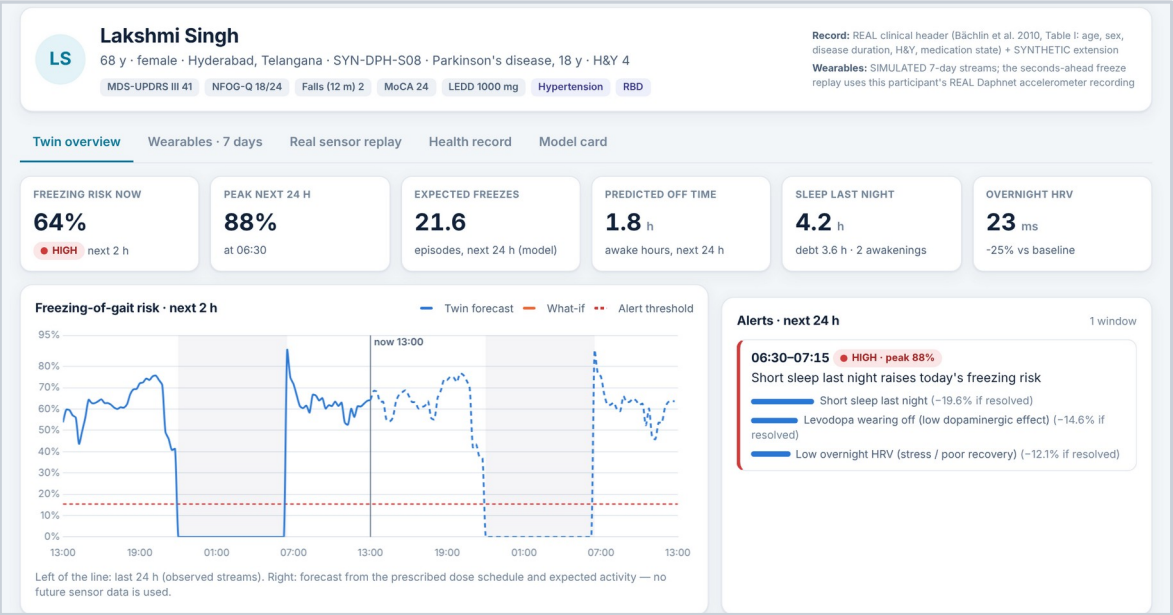
- Synthetic FHIR R4 record (SNOMED CT, LOINC), tagged SYNTHETIC; real Daphnet clinical header for 10 people
- Levodopa PK/PD: dopamine effect and ON/OFF state
- Simulated 7-day wearables + real Daphnet sensors
- Model A: freeze within 3 s, AUC 0.76 (0.72–0.83), real data, leave-one-person-out
- Model B: freeze within 2 h, AUC 0.806, synthetic cohort
- What-if re-simulation; game session synced at the end of Level 3

PROPOSED, NOT CLAIMED

- Game measures as inputs to Models A and B: needs clinician-labelled game sessions
- Personal PK/PD fitted to real patients: today it uses published population values
- A validated, predictive twin of an individual
- Model B accuracy reflects the simulator, not clinical accuracy

Digital twins in health [11] · Daphnet clinical header [9] · Levodopa PK: Hayashi 2025 · Closest published pre-freeze benchmark (AUC 0.76): Palmerini 2017 [13]

Clinical dashboard and assessment outputs



Twin overview: 2-hour freezing risk, alerts with drivers, ON/OFF curve (synthetic patient)

Domain	What the clinician sees	Source
Voice	Sustained vowel, pa-ta-ka, rhythm, speech rate	Mic · game
Visual tracking	On-target %, error, re-acquisition, reaction	Webcam · game
Spiral / fine motor	Radial deviation, tremor-band proxy, smoothness	Game · wrist IMU*
Cognition	Accuracy and response time per task	Game
Pose and movement	Peak arm elevation, left vs right arm, pose similarity	Webcam
Gait	Cadence, step variability and alternation, stop/go latency; freeze alerts	Webcam · ankle IMU*
Questionnaire	Six domains, as reported, never scored	Game
Trend	Game sessions listed per patient, beside the ON/OFF curve	Twin record

* when the wearable is connected

AFTER THE GAME

Level 3 twin report	→	OPEN CLINICIAN	→	Opens on the
6 charts · session syncs		DASHBOARD button		player's own twin

Also F2 anywhere, and CLINICIAN DASHBOARD on the title screen, Three Lands and pause menu. Live from the twin service; otherwise an offline snapshot. Tabs: overview, what-if, sensor replay, game sessions, record, model card.

How it fits together

Game metrics and the questionnaire are stored on the twin and shown beside the levodopa ON/OFF curve. The clinician reads them together with the risk forecast; they do not yet feed the models. Model A warns 62% (148/237) freezes with a 4.6 s median lead.

Dashboard values are synthetic patient data (Daphnet-anchored twin). Model results come from ParkiScanML/reports; AUC with 95% subject-bootstrap CI.

Home use: assessment today, rehabilitation as an extension

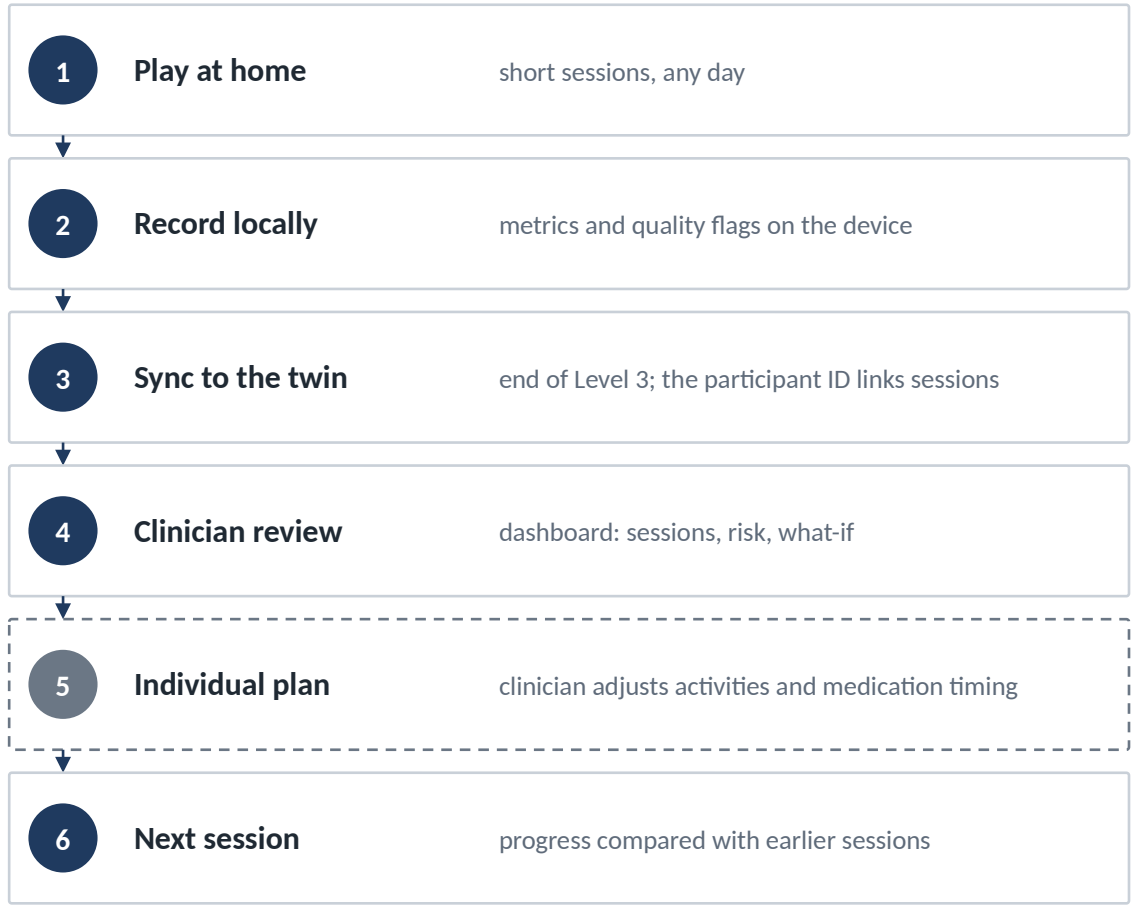
TRACK 1 · ASSESSMENT AND MONITORING (BUILT)

- Anyone can play; the output is measures and flags, never a diagnosis.
- Short sessions, seated mode for Level 3, large targets, skip on every task.
- Each session is time-stamped, so measures line up with the medication ON/OFF curve.
- The clinician reviews sessions on the dashboard before the next visit.

TRACK 2 · REHABILITATION FOR DIAGNOSED USERS (PROPOSED)

- Only for people already diagnosed; activities chosen by their clinician.
- Candidates already in the game: rhythm clapping and cueing, mirrored arm movements, marching on the spot, steering tasks, cognitive and reaction tasks.
- Evidence: VR and exergames in PD are feasible and safe; results are similar to conventional exercise, low certainty [4,5]. Rhythmic cueing improves gait in PD [12].
- No therapeutic claim is made for Neuroverse; it needs its own trial.

HOME-TO-CLINIC LOOP



[4] Dockx 2016 · [5] Garcia-Agundez 2019 · [12] Ghai 2018 (rhythmic auditory cueing meta-analysis) · Evidence is for serious games in general, not for Neuroverse.

Hackathon requirements and references

Requirement (challenge)	Neuroverse	Where
Twin of one patient, one adverse event	Parkinson's patient; freezing of gait	Twin service
Static data: synthetic EHR	FHIR R4, SNOMED CT, LOINC, tagged SYNTHETIC	twin/ehr.py
Dynamic data: wearables / IoT	Real Daphnet IMUs; simulated steps, HR, sleep, HRV; ESP32 path	Daphnet, firmware
Fuse and predict	Model A AUC 0.76; Model B AUC 0.806 (record only 0.779)	reports/
Doctor dashboard	In the game (F2, post-game) and browser; what-if	Game, /dashboard
Sandbox data rules	Synthetic or anonymised only; processed locally	Whole project
Submission package	GitHub, Apache-2.0, architecture, deck; 20-min video in preparation	Repository

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Datasets: Hausdorff 2000 J Appl Physiol 88:2045 · Little 2007 BioMed Eng OnLine 6:23 · Sakar 2013 IEEE JBHI 17:828 · Sakar 2018 (UCI) · Tsanas 2010 IEEE TBME 57:884 · Isenkul 2014 (UCI) · ET_PD_VIS and TMT-PD, Mendeley Data 2026.